Sri Chaitanya ACADEMY | Infinity Learn



JEE FINAL TEST SERIES

JEE Advanced Full Syllabus Test - 9 [2024 P2]

Test Date: 04th May 2025

INSTRUCTIONS

- Use the OMR sheet provided separately for answering the questions.
- Blank spaces are provided within this booklet for rough work.
- Write your roll number in the space provided on the front cover of this Question Paper
- You are allowed to take away the Question Paper at the end of the examination.
- The OMR will be collected by the invigilator at the end of the examination.
- Do not tamper with or mutilate the OMR. Do not use the OMR for rough work.
- Write your name, roll number and sign with pen in the space provided for this purpose on the OMR.
- Do not write any of these details anywhere else on the OMR. Darken the appropriate bubble under each digit of your roll number.
- Use a BLACK BALL POINT PEN to darken the bubbles on the OMR>
- The OMR is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.

JEE-ADVANCED-2024-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with Integer Answer Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Paragraph with stem NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 21)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 22 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 25 – 30)	Questions with Integer Answer Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Paragraph with stem NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 37)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 38 – 41)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 42 – 47)	Questions with Integer Answer Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Paragraph with stem NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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1. The mean square deviation of set of n observations $x_1, x_2, x_3 \ldots\ldots x_n$ about a point c is defined as $\frac{1}{n}\sum_{i=1}^{n}(x_i - c)^2$. The mean square deviation about -2 and are 2 are 18 and 10 respectively, then the standard deviation of this set of observations is equal to

A) 3 **B)** 2 **C)** 1 **D)** 4

2. Let R be real numbers and $f: R \to R$ such that $\forall x, y \in R, |f(x) - f(y)| \le |x - y|^{100}$ and $f(100) = 1$, then find the area bounded by curve y=min.$\{f(x), x^2, e^{2-x}\}$, x-axis and line $x = 4$. in sq.unit

A) $\frac{7}{3}$ **B)** $\frac{7}{3} - \frac{1}{e^2}$ **C)** $\frac{4}{3} - \frac{1}{e^2}$ **D)** $\frac{7}{3} + \frac{1}{e^2}$

3. For the real number x, let $[x]$ denote the greatest integer less than or equal to x . Let $f: R \to R$ be defined as $f(x) = 2x + \left[\frac{x(x^2-1)}{4(x^4 - x^2 + 1)} + \frac{1}{8}\right] + [x] + \sin x.\cos x$, then f is

A) one – one but not onto **B)** onto but not one-one

C) both one –one and onto **D)** neither one –one nor onto

4. Given $f'(0) = 1$ and if $f: (-1,1) \to (-1,1)$ be a differentiable function defined such that $f(x) + f(y) = f\left(\frac{x\sqrt{1-y^4} + y\sqrt{1-x^4}}{1 + x^2y^2}\right) \ \forall \ x, y \in (-1,1)$ then $f'\left(\frac{1}{\sqrt{3}}\right) =$

A) $\frac{3}{2\sqrt{2}}$ **B)** $\frac{1}{\sqrt{2}}$ **C)** $\frac{1}{2}$ **D)** $\frac{1}{3}$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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5. If $\int \frac{\sin 2x}{(3+4\cos x)^3}dx = \frac{a+b\cos x}{c(3+4\cos x)^d}+e$ where, a,b,c,d are positive integers and e is arbitrary constant then

A) The value of a + b is 11 **B)** The value of a + b is 6

C) The value of c +d is 18 **D)** The value of c – d is 14

6. Let f be a twice differentiable function defined in $[-4,4]$ such that $f'(-4)=10$, $f'(4)=-14$ and $f''(x)\geq -3\ \forall\ x\in[-4,4]$. If $g(x)=\int_0^x f(t)dt$ and $f(0)=0$, which of the following options is/are correct ?

A) $g(x)$ increases in (-4,0) **B)** $g(x)$ decreases in (0,4)

C) graph of $g(x)$ is concave down in (0,4) **D)** $f(x)$ has maximum value at $x=2$

7. If A and B are square matrices of order '3' such that Aadj(2B) =16 I, and det (B) =2 then which of the following is/are true (Where adj(X) denotes adjoint of matrix (X), X^{-1} denotes inverse of matrix (X) and det (X) denotes determinant value of matrix (X).

A) $\left(A^{-1}(adjB)\right)^{-1}=4A^2$ **B)** $\left(A^{-1}(adjB)\right)^{-1}=B^2$ **C)** $B(adjA)=8I$ **D)** $B(adjA)=4I$

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

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8. In ΔABC, the co-ordinates of the vertex A are (4,-1) and lines $x-y-1=0$ and $2x-y=3$ are internal bisectors of angles B and C . Then radius of in-circle of ΔABC is $\frac{k}{\sqrt{5}}$ where k =

9. Let the variable circle $S: x^2+y^2-ax-by+c=0$ intersects the pair of straight lines $xy-4x-3y+12=0$ orthogonally and the circle lies in the first quadrant. If S touches the circle $S_1:(x-3)^2+y^2=r^2,\ r\in N$, then the number of all possible value of r is .

10. A tangent is drawn to the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ to cut the ellipse $\frac{x^2}{c^2}+\frac{y^2}{d^2}=1$ at the points P and Q. If the tangents at P and Q to the ellipse $\frac{x^2}{c^2}+\frac{y^2}{d^2}=1$ intersect at right angles, then the value of $\frac{16}{c^2}+\frac{9}{d^2}$ is equal to

11. Let Z_1, Z_2, Z_3 be complex numbers with $|z_1|=|z_2|=|z_3|=2$. Then the maximum of $\left|Z_1Z_2+Z_2Z_3-Z_1Z_3-Z_2^2\right|+\left|Z_2Z_3+Z_3Z_1-Z_2Z_1-Z_3^2\right|+\left|Z_3Z_1+Z_1Z_2-Z_2Z_3-Z_1^2\right|$ is

12. In an Arithmetic progression with first term 1. Consider any even number of terms from the beginning, then the sum of the first half of these terms and the sum of second half of the remaining terms are in the constant ratio, then common difference is ……..(common difference $d>0$)

13. In a sequence of coin tosses, one can keep a record of instances in which a tail is immediately followed by a head , a head is immediately followed by a head, and etc. We denote these by TH,HH and etc . For example, in the sequence TTTHHTHTTTHHTTH OF 15 coin tosses we observe that there are two HH, three HT, four TH, and five TT subsequences. If the number of different sequences of 15 coin tosses will contain exactly two HH, three HT, four TH, and five TT subsequences is k then value of $\frac{k}{80}$ is ?

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

Paragraph-I

Let $f(x)=x^4-8x^3+22x^2-24x\ \forall\ x\in R$ and function $g(x)$ is defined as:

$$g(x)=\begin{cases}\min\{f(t):0\le t\le x\}; & 0\le x<2\\ \max\{f(t):2<t\le x\}; & 2\le x\le 5\end{cases}$$

On the basis of given definition of $f(x)$ and $g(x)$ answer the following questions:

14. If it is possible that $g(x)$ in $(0,5)$ is non- differentiable at P (finite number) points, then P=

15. Value of $g'\left(\frac{3}{2}\right)$ is equal to :

Paragraph-II

In a tetrahedron ABCD, the face ABC is a regular (equilateral) triangle and the face DBC is perpendicular to it . $\underline{|DAC} = \frac{\pi}{3}, |AD| = 6cm$ and the angle between $\overline{AD}$ and $\overline{BC}$ is $arc\left(\cos\left(\frac{1}{4}\right)\right)$ If A is origin, $\overline{AD} = \overline{d}.\overline{AB} = \overline{b}, \overline{AC} = \overline{c}$

16. The value of $|b| + |c| =$

17. Volume of tetrahedron ABCD =

PHYSICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
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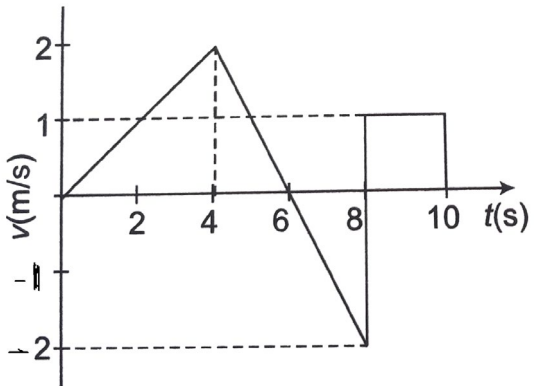
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18. The velocity v of a particle at time t is given by $v = at + \frac{b}{t+c}$, where a, b and c are constants . The dimensions of a, b and c are, respectively ,

A) $[LT^{-2}], [L] \text{and} [T]$ **B)** $[L^2], [T] \text{and} [LT^2]$

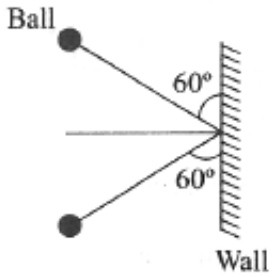
C) $[LT^{-2}], [LT] \text{and} [L]$ **D)** $[L], [LT] \text{and} [T^2]$

19. The velocity –time graph of a body moving in a straight line is shown. The displacement of the body in 10 s is .



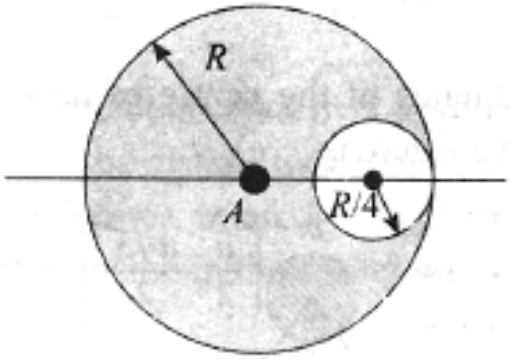
A) 4 m **B)** 6 m **C)** 8 m **D)** 10 m

20. A ball of mass 3 kg moving with a speed of 100 m/s , strikers a wall at an angle 60^0 as shown in the figure. The ball rebounds at the same speed and remains in contact with the ball for 0.2 s. The force exerted by the ball on the wall is



A) $1500\sqrt{3}N$ **B)** $1500\ N$ **C)** $300\sqrt{3}N$ **D)** $300N$

21. The centre of mass of the shaded portion of the disc is (The mass is uniformly distributed in the shaded portion)



A) $\frac{R}{20}$ to the left of A **B)** $\frac{R}{12}$ to the left of A

C) $\frac{R}{20}$ to the right of A **D)** $\frac{R}{12}$ to the right of A

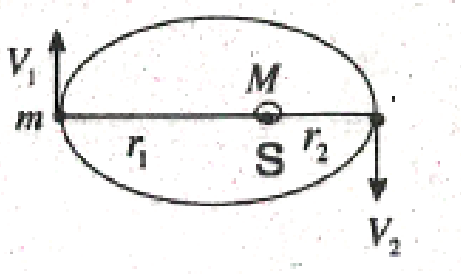
SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
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22. A planet of mass M moves along an ellipse around then sun, so that its maximum and minimum distances from the sun are r_1 and r_2 respectively. Then



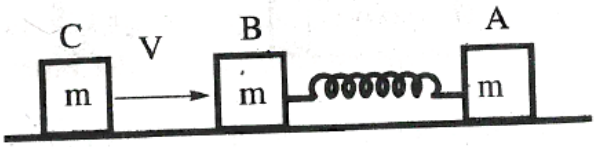
A) $V_1 r_1 = V_2 r_2$

B) Total energy of plant is always a constant

C) Angular momentum of plant is $m\sqrt{\frac{2GM(r_1 r_2)}{(r_1 + r_2)}}$

D) $V_2 r_1 = V_1 r_2$

23. Two blocks A and B each of mass m are connected by a massless spring natural length ' l' and spring constant K. The blocks are initially resting on a smooth horizontal floor with the spring at its natural length as shown. A third identical block C also of mass m, moves on the floor with a speed v along line joining A and B and collides elastically with B, then



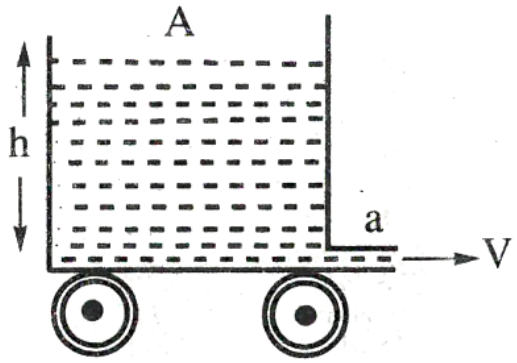
A) The K.E of the A-B system, at the maximum compression of the spring is zero .

B) The K..E of the system, at the maximum compression of the spring is $mv^2/4$.

C) The maximum compression of the spring is $v\sqrt{\frac{m}{K}}$

D) The maximum compression of the spring is $v\sqrt{\frac{m}{2K}}$

24. A container having a hole at the bottom is free to move on a frictionless horizontal surface. As the liquid comes out. Container moves in backward direction with an acceleration α and finally acquired velocity v (When all liquid drains out) Neglect the mass of container. The correct options are



A) α is independent of height of liquid in vessel

B) α depends only A and a and g

C) v depends upon α and h

D) v independent of α

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
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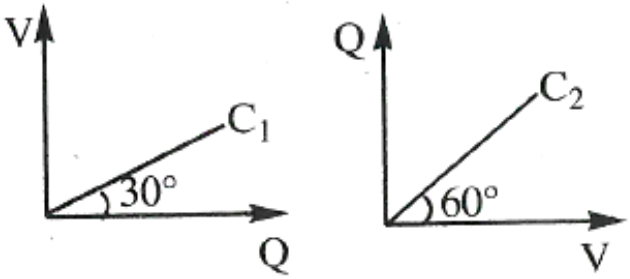
Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases..

25. Tuning fork A of frequency 258 Hz gives 8 beats with a tuning fork B. When the tuning fork A is filed and again A and B sounded the number of beats decreases. The frequency of B is ……

26. A heavy wheel of radius 20 cm and weight 10 kg is to be dragged over a step of height 10 cm, by a horizontal force F applied at the centre of the wheel . The minimum value of " F " is (in kgwt) ……..

27. The refractive indices of crown glass prism for C,D and F lines are 1.527 , 1.530 and 1.535 respectively. Then the dispersive power of the crown glass prism is ……. (round off nearest integer)

28. Two different capacitors with capacity C_1 and C_2 are being charged. Graphs for C_1 and C_2 are as follow. Find the value of C_1 / C_2.



29. The speeds of 5 molecules of gas (in arbitrary units) are as follows 2,3,4,5,6. The root mean square speed for these molecules is ……..

30. when temperature of a gas is 20^0C and pressure is changed from $P_1 = 1.01\times10^5\,Pa$ to $P_2 = 1.165\times10^5\,Pa$ then volume changed by 10 %. The bulk modulus is $n\times10^5$ Pa then n is...

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

Paragraph-I

The thermal coefficient of resistivity is given $\alpha = \frac{1}{\rho}\frac{dp}{dT}$ where ρ is resistivity at a temperature T. α is not constant, rather varies as $\alpha = \frac{-n}{T}$, where T is temperature in Kelvin and n is a constant . For carbon $\alpha = -0.0005K^{-1}$ and $\rho = 3.5\times10^{-5}(\Omega - m)$ at 27^0C

31. The value of $\frac{d\rho}{dT}$ is $-x\times10^{-8}\Omega m/K$, find x (round off nearest integer)

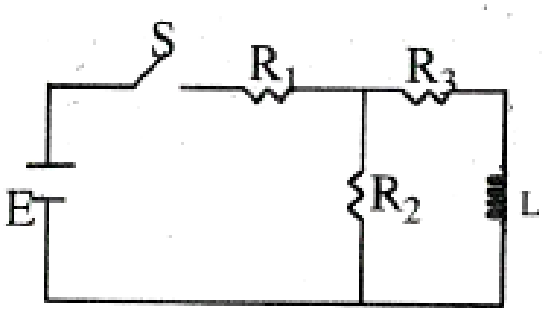
32. Find the value of ' n' for carbon (round off nearest integer)

Paragraph-II

$E=50V$, $R_1=10\Omega$

$R_2=20\Omega$, $R_3=30\Omega$

and L=2mH



33. After the steady state is attained , current through R_1 immediately after S is reopened is

34. After the steady state is attained, current through R_2 immediately after S is reopened is nearly (round off nearest integer)

CHEMISTRY **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

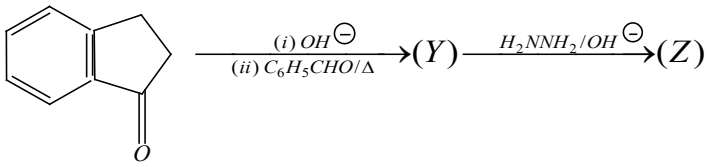
This section contains **FOUR (04)** questions.

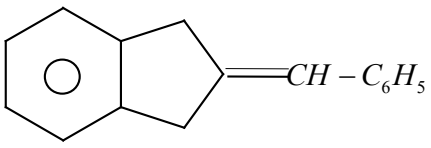
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
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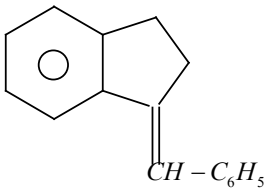
Full Marks : +3 If **ONLY** the correct option is chosen;
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Negative Marks : −1 In all other cases

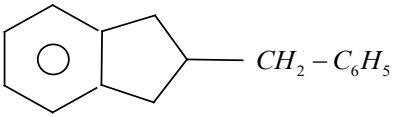
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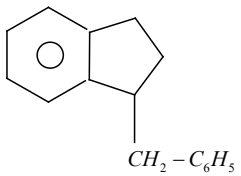
35. In the given reaction sequence compound (Z) is:



A) 

B) 

C) 

D) 

36. $H_2C_2O_4 \xrightarrow{\Delta}$ + Gas A + Gas B + liquid C

Gas A burns with blue flame and is oxidized to gas B

Gas $A + Cl_2 \longrightarrow D \xrightarrow{NH_3/\Delta} E$ Identify A,B,C and E

A) $CO_2, CO, H_2O\ HCONH_2$ **B)** $CO, CO_2, COCl_2, HCONH_2$

C) $CO, CO_2, H_2O\ NH_2CONH_2$ **D)** $CO, CO_2, H_2O, COCl_2$

37. A reaction with respect to 'X' is zero order till the concentration is reduced to half of initial concentration then the reaction becomes first order with respect to it. If the value of rate constant for the zero order and first order equal to K [in magnitude]. Then find the time taken to reduce concentration of X to $\frac{1}{16}$ the of its original concentration X_0

A) $\frac{1}{2k} + \frac{3\ell n 2}{k}$ **B)** $\frac{X_0}{2k} + \frac{3\ell n 2}{k}$ **C)** $\frac{X_O}{2k} + \frac{4\ell n 2}{k}$ **D)** $\frac{1}{2k} + \frac{4\ell n 2}{k}$

38. Consider the following cell

$Ag_{(s)} \left| AgBr_{(s)} \right| \left| Br^-_{(aq)} \right| \left| cl^-_{(aq)} \right| \left| AgCl_{(s)} \right| Ag_{(s)}$ at 25^0C

K_{SP} of $AgBr = 5 \times 10^{-13}\ m^2$

K_{SP} of $AgCl = 1 \times 10^{-10}$

For what ratio of concerns of Br^- and Cl^- would the e.m.f of the cell be zero

A) 1 : 200 **B)** 1 : 100 **C)** 1 : 500 **D)** 200 : 1

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

39. Which of the following pairs form same osazone?

A) Glucose, Fructose **B)** Glucose, Manose

C) Ribose, Arabinose **D)** Mannose,fructose

40. Both H_2O_2 and O_3 act as oxidizing agents. The oxidizing action in which they are similar is

A) PbS to $PbSO_4$ **B)** Fe^{+2} *to* Fe^{+3}

C) $K_4\left[Fe(CN)_6\right]$ *to* $K_3\left[Fe(CN)_6\right]$ **D)** Decolourising acidified $KMnO_4$

41. Which of the following are correct statements

A) Spontaneous adsorption of gases on solid surface is an exothermic process as entropy decreases during adsorption.

B) Formation of micelles takes place when temperature is below kraft temperature (T_k) and concentration above cmc value (cmc = critical micelle concentration)

C) Longer the length of hydrophobic chain smaller will be the cmc value

D) According hardy – schuzle rule the coagulation value of Fe^{+3} ion will be more than Ba^{+2} or Na^{+}

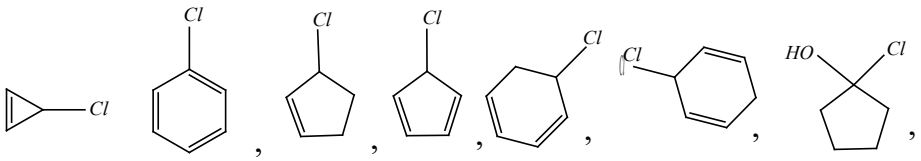
SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

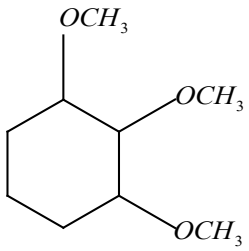
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42. How many of the following halides gives white precipitate with aqueous $AgNO_3$ solution?



$H_2C = CH - Cl$, $CH_2 = CH - CH_2 - Cl$, $ph - CH_2 - Cl$

43. Moles of HI consumed for complete reaction with one mole of the following compound is_____



44. The average degree of freedom permolecule for a gas is 6. The gas performs 25J of work. When it expands at constant pressure the amount of heat absorbed is $4x^2J$ the value of x

45. The solubility of sliver acetate in pure water at 25^0C 8.35 gm/lit and 61.8 gms/lit in acetic acid solution of pH = 3 the dissociation constant of acetic acid is $y \times 10^{-x}$

Find the value x.

46. For a reversible reaction $A \underset{}{\overset{k_f}{\rightleftharpoons}} B$

Rate constant K_1 [Forward] $= 10^{15} e^{\frac{-2000}{T}}$

And k_2 (back ward) $= 10^{12} e^{-2000/T}$

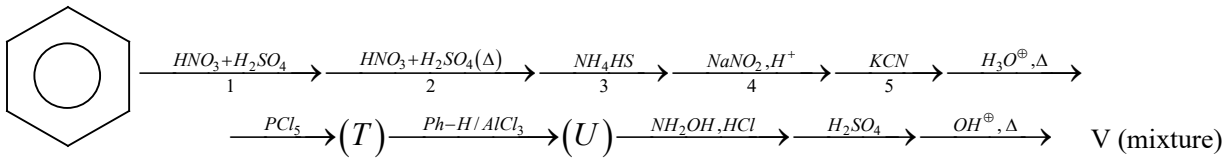
What is the value of $-\frac{\Delta G^{O}}{2.303\,RT}$

47. Consider the following reagents acidified $K_2Cr_2O_7$ Alkaline $KMnO_4, CuSO_4, H_2O_2, Cl_2, O_3, FeCl_3\ HNO_3\ and\ Na_2S_2O_3$. The total number of reagents that can oxidize aqueous iodide to iodine is.

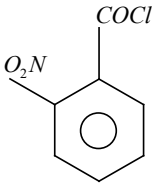
SECTION-IV
(PARAGRAPH QUESTIONS)

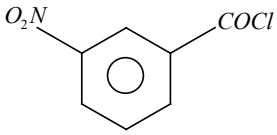
- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

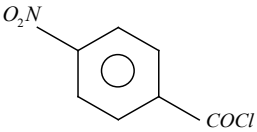
Paragraph-I

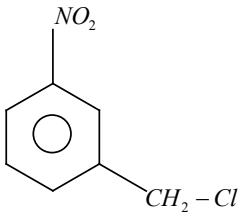


48. The product is T is

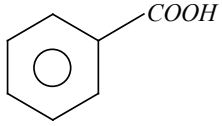
1) 

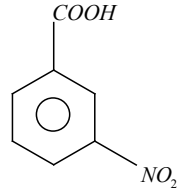
2) 

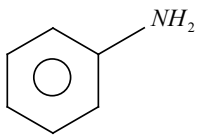
3) 

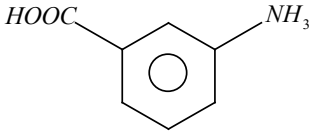
4) 

49. The product mixture 'V' does not have.

1) 

2) 

3) 

4) 

Paragraph-II

Five litres of 16.8 volume of H_2O_2 solution of density 1.051 gm/m1 is taken in a 50 lit vessel and the vessel is sealed

50. The molality of the solution is

51. If H_2O_2 present in the solution get decomposed completely then total mass of water finally present in the vessel should be (in grams)